

MA 120: Introduction to Computing System and Databases

Course Code: MA 120

Course Name: Introduction to Computing System and Databases

L-T-P-C: 3-0-2-4

Students intended for: B.Tech Maths and Computing

Elective or Core: Core

Prerequisites: Basic Knowledge of Programming and Data Structures

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Introduction to Computer Organization and Data Representation** [12 Lectures]

- Data Representation: Number systems (binary, octal, hexadecimal), signed numbers, floating-point representation.
- Machine Code: Introduction to machine language and assembly language, instruction formats, opcode and operands, addressing modes.
- Computer Arithmetic: Addition, subtraction, multiplication, and division, fixed-point and floating-point arithmetic, overflow and underflow.
- Memory Organization and Management: Memory hierarchy, cache design, RAM, ROM, memory addressing, memory paging and segmentation.

- **Introduction to Operating Systems:** [15 Lectures]

- Responsibilities of Operating Systems: Basic functions of an operating system, user-interface, and system-call interface.
- Process & Thread Management:
 - * Process model and states: Process life cycle, process control block, context switching.
 - * Thread models: Difference between processes and threads, multi-threading.
 - * User/Kernel Level Threads: Thread management and scheduling at user and kernel levels.
- Process Creation and Termination: Forking and exec operations, process termination, exit statuses.
- Process Synchronization: Critical section problem, semaphores, mutex, condition variables.
- Inter-Process Communication (IPC): Message passing, shared memory, pipes, and sockets.
- Process Scheduling: Scheduling algorithms such as FCFS, SJF, Round Robin, Priority Scheduling.
- File Systems: File structure, file access methods, file system implementation, file allocation strategies.
- • Memory Organization: Primary, secondary memory, memory blocks, paging, segmentation.
- • Virtual Memory: Address translation, page tables, page faults, thrashing.

- **Introduction to Databases:** [15 Lectures]

- Information Modeling: Conceptual data models, Entity-relationship diagrams, Normal Forms .

- ACID Properties and Transactions: Understanding Atomicity, Consistency, Isolation, Durability, and the role of transactions in databases. Serializability, concurrency control, and lock-based mechanisms like Two-Phase Locking (2PL) and deadlock detection.
- Data Manipulation Language (SQL)
 - * SQL Basics: The basic SQL commands include SELECT, INSERT, UPDATE, and DELETE for querying and modifying data.
 - * Advanced SQL: This covers more complex SQL operations such as joins, subqueries, aggregate functions, indexing, and views for efficient data manipulation.

Laboratory/practical/tutorial Modules:

- Based on the contents discussed in class lab will consist of the following:
 - MIPS Programming (QtSPIM)
 - * Writing, running, and debugging MIPS assembly programs.
 - Shell Scripting
 - * Writing and executing bash scripts for task automation.
 - Multiprocessing
 - * Implementing threads, synchronization, and handling race conditions.
 - Multithreading
 - * Implementing threads, synchronization, and handling race conditions.
 - SQL

Text books:

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, **Operating System Concepts**, Wiley.
2. David A. Patterson and John L. Hennessy, **Computer Organization and Design**, Morgan Kaufmann.
3. Raghu Ramakrishnan and Johannes Gehrke, **Database Management Systems**, McGraw-Hill.

References:

1. Andrew S. Tanenbaum, **Modern Operating Systems**, Pearson.
2. Hector Garcia-Molina, Jeff Ullman, Jennifer Widom, **Database Systems: The Complete Book**, Pearson.
3. Andrew S. Tanenbaum, **Structured Computer Organization**, Pearson.